MATH 350-2 Advanced Calculus

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Outline

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1. Real Analysis Lecture 18
   - Uniform continuity
   - Infinite series
   - Infinite products

Uniform continuity

Definition

Let (S, d_S) and (T, d_T) be metric spaces and $A \subseteq S$. We say that a function $f : A \to T$ is **uniformly continuous** on A if for all $\epsilon > 0$ there exists $\delta > 0$ such that

$$d_S(x, y) < \delta \Rightarrow d_T(f(x), f(y)) < \epsilon$$

for all $x, y \in A$.

- f continuous on A means f is continuous at every $a \in A$
- δ can depend on ϵ and a (not on x)
- with uniform continuity, δ can't depend on a ...

Examples

- x is uniformly continuous on $\mathbb{R}$
- x^2 is NOT uniformly continuous on $\mathbb{R}$
- $1/x$ is continuous on $(0, 1)$
- $1/x$ is NOT uniformly continuous on $(0, 1)$

Relationship with compactness

Theorem (Heine)

Let (S, d_S) and (T, d_T) be metric spaces and $A \subseteq S$ be compact. Suppose that $f : A \to T$ is continuous on A, then f is uniformly continuous on A.

Outline

Basic definition

Definition

An **infinite series** is an ordererd pair of sequences $(\{a_n\}, \{s_n\})$ where

$$s_n = a_1 + a_2 + \dots + a_n.$$

The sequence $\{s_n\}$ is called the **sequence of partial sums** and s_n is called the n'th partial sum. We say the series converges or diverges if $\{s_n\}$ converges or diverges.

Notation:

$$a_1 + a_2 + a_3 + \dots \quad \text{or} \quad \sum_{n=1}^{\infty} a_n.$$

We write

$$\sum_{n=1}^{\infty} a_n = L$$

Examples

$$\sum_{n=1}^{\infty} \frac{1}{n^2+n} = 1.$$

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Telescoping:

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Telescoping:

$$\begin{aligned} s_n &= \frac{1}{1^2+1} + \frac{1}{2^2+2} + \frac{1}{3^2+3} + \cdots + \frac{1}{n^2+n} \\ &= \left(\frac{1}{1} - \frac{1}{2}\right) + \left(\frac{1}{2} - \frac{1}{3}\right) + \cdots + \left(\frac{1}{n} - \frac{1}{n+1}\right) \\ &= 1 - \frac{1}{n+1}. \end{aligned}$$

Examples

$$\sum_{n=1}^{\infty} r^n = \frac{r}{1-r}, \quad |r| < 1.$$

Examples

Geometric:

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Geometric:

$$s_n = r + r^2 + r^3 + \cdots + r^n = r(1 + r + r^2 + \cdots + r^{n-1}) = r(1 + s_{n-1})$$

Examples

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Geometric:

$$s_n = r + r^2 + r^3 + \cdots + r^n = r(1 + r + r^2 + \cdots + r^{n-1}) = r(1 + s_{n-1})$$

$$s_n = r + r^2 + r^3 + \cdots + r^n = (r + r^2 + \cdots + r^{n-1}) + r^n = s_{n-1} + r^n$$

Examples

$$\sum_{n=1}^{\infty} r^n = \frac{r}{1-r}, \quad |r| < 1.$$

Geometric:

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$$s_n = \frac{r - r^{n+1}}{1 - r}$$

Examples

$$\sum_{n=1}^{\infty}(-1)^{n+1}\frac{1}{n} = \ln(2)$$

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Taylor series:

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Taylor series:

$$\ln(1-x) = -x - \frac{1}{2}x^2 - \frac{1}{3}x^3 - \frac{1}{4}x^4 - \dots$$

Examples

$$\sum_{n=1}^{\infty}(-1)^{n+1}\frac{1}{n} = \ln(2)$$

Taylor series:

$$\ln(1-x) = -x - \frac{1}{2}x^2 - \frac{1}{3}x^3 - \frac{1}{4}x^4 - \dots$$

Converges for $-1 \leq x < 1$.

Tests for Convergence

Theorem (Cauchy condition)

The series $\sum a_n$ converges if and only if for all $\epsilon > 0$ there exists $N \in \mathbb{Z}_+$ such that $n \geq N$ implies

$$|a_n + a_{n+1} + \cdots + a_{n+p}| < \epsilon$$

for all $p \in \mathbb{Z}_+$.

Tests for Convergence

Definition

An **alternating series** is a series of the form $\sum(-1)^{n+1}a_n$ with $a_n \geq 0$ for all n.

Theorem (Alternating Series Test)

If a_n is a decreasing sequence converging to 0, then $\sum(-1)^{n+1}a_n$ converges to a real number s and

$$|s - s_n| < a_{n+1} \quad \forall\, n \in \mathbb{Z}_+.$$

Tests for Convergence

Definition

We say $\sum a_n$ is **absolutely convergent** if $\sum |a_n|$ converges. We say $\sum a_n$ is **conditionally convergent** if it is convergent but not absolutely convergent.

Theorem (Absolute Convergence Test)

If a series is absolutely convergent, then it is convergent.

Tests for Convergence

Theorem (Comparison Test)

If $a_n > 0$ and $b_n > 0$ and there exists $c > 0$ and $N \in \mathbb{Z}_+$ with $a_n < cb_n$ for all $n \geq N$ then

$$\sum b_n \text{ converges} \quad \Rightarrow \quad \sum a_n \text{ converges}$$

Tests for Convergence

Theorem (Limit Comparison Test)

If $a_n > 0$ and $b_n > 0$ and suppose that $\lim a_n/b_n = c > 0$. Then

$$\sum b_n \text{ converges} \quad \Leftrightarrow \quad \sum a_n \text{ converges}$$

Limit infimum/limit supremum

Definition

Let a_n be a sequence. Then

$$\liminf a_n = \lim_n \inf\{a_k : k \geq n\}$$

$$\limsup a_n = \lim_n \sup\{a_k : k \geq n\}$$

Theorem

If a_n is bounded, then $\limsup a_n$ and $\liminf a_n$ exist.

More generally, we write $\liminf a_n = \pm\infty$ and $\limsup a_n = \pm\infty$.

Tests for Convergence

Theorem (Ratio Test)

Let $r = \liminf |a_{n+1}/a_n|$ *and* $R = \limsup |a_{n+1}/a_n|$.

(a) *if* $R < 1$ *then the series is absolutely convergent*
(b) *if* $r > 1$ *then the series is divergent*
(c) *if* $r \leq 1 \leq R$ *the test is inconclusive*

Tests for Convergence

Theorem (Root Test)

Let $\rho = \limsup \sqrt[n]{|a_n|}$.

(a) *if* $\rho < 1$ *then the series is absolutely convergent*
(b) *if* $\rho > 1$ *then the series is divergent*
(c) *if* $\rho = 1$ *the test is inconclusive*

Outline

Basic definition

Definition

An **infinite product** is an ordererd pair of sequences $(\{a_n\}, \{p_n\})$ where

$$p_n = a_1 a_2 \dots a_n.$$

The sequence $\{p_n\}$ is called the **sequence of partial products** and p_n is called the n'th partial product. We say the series converges or diverges if $\{p_n\}$ converges or diverges.

Notation:

$$a_1 a_2 a_3 \dots \quad \text{or} \quad \prod_{n=1}^{\infty} a_n.$$

We write

$$\prod_{n=1}^{\infty} a_n = L$$

Examples

Weierstrass product of sine

$$\sin(\pi x) = \pi z \prod_{n=1}^{\infty} \left(1 - \frac{x^2}{n^2}\right)$$

- first thought of by Euler.
- can be used to prove

$$\sum \frac{1}{n^2} = \frac{\pi^2}{6}.$$

Examples

Let $p_1 = 2, p_2 = 3, p_3 = 5, \dots$ be the prime integers.
The infinite product

$$\zeta(s) = \prod_{n=1}^{\infty} \frac{1}{1 - p_n^{-s}}$$

converges for $s > 1$.
This product is called the **Riemann zeta function.**

Examples

Theorem

For any real number $s > 1$,

$$\zeta(s) = \sum_{n=1}^{\infty} \frac{1}{n^s}.$$

Therefore

$$\frac{1}{(1 - 1/2^2)(1 - 1/3^2)(1 - 1/5^2)(1 - 1/7^2)(1 - 1/11^2)\dots} = \frac{\pi^2}{6}.$$

Relation to series

Theorem

Assume $a_n > 0$ for all n. Then $\prod(1 + a_n)$ converges if and only if $\sum a_n$ converges.

Relation to series

Definition

We say that a product $\prod(1+a_n)$ **converges absolutely** if $\prod(1+|a_n|)$ converges.

Theorem

If $\prod(1+a_n)$ converges absolutely (equiv. $\sum a_n$ converges absolutely), then $\prod(1+a_n)$ converges.